The Water Cycle

The water cycle, also known as the hydrological cycle, describes the continuous movement of water on, above, and below the surface of the Earth. This cycle is driven by solar energy and gravity.

Key Vocabulary

Evaporation - The process by which water changes from liquid to gas or vapor. This occurs when the sun heats water in rivers, lakes, or oceans.

Condensation - The process by which water vapor in the air is changed into liquid water. This is crucial to the formation of clouds.

Precipitation - Water released from clouds in the form of rain, sleet, snow, or hail. It is the primary way that water returns to Earth's surface.

Transpiration - The process by which plants release water vapor into the atmosphere through their leaves. This accounts for approximately 10% of atmospheric moisture.

Infiltration - The process by which water on the ground surface enters the soil. Water that infiltrates can become groundwater.

Runoff - Water that flows over the land surface and into streams, rivers, and eventually the ocean. This occurs when the ground is saturated or impermeable.

Groundwater - Water stored beneath Earth's surface in soil and rock formations called aquifers.

Aquifer - An underground layer of water-bearing permeable rock or sediment from which groundwater can be extracted.

Percolation - The movement of water through soil and porous rock, eventually reaching groundwater reservoirs.

Sublimation - The transition of water directly from solid (ice/snow) to gas (water vapor) without passing through the liquid phase.

Stages of the Water Cycle

1. Evaporation and Transpiration

Solar radiation heats water in oceans, lakes, and rivers, causing it to evaporate into water vapor. Plants also release water vapor through transpiration. Together, these processes are called evapotranspiration.

2. Condensation

As water vapor rises into the atmosphere, it cools and condenses around tiny particles to form clouds. This process releases heat energy into the atmosphere.

3. Precipitation

When cloud droplets combine and become heavy enough, they fall to Earth as precipitation. The form depends on atmospheric temperature - rain, snow, sleet, or hail.

4. Collection and Infiltration

Precipitation that reaches Earth's surface either: - Runs off into bodies of water (rivers, lakes, oceans) - Infiltrates into the ground to become groundwater - Is absorbed by plants or evaporates back into the atmosphere

Important Concepts

Residence Time - The average time a water molecule spends in a particular reservoir. Ocean water has a residence time of about 3,000 years, while atmospheric water vapor has a residence time of only about 9 days.

Water Distribution - Approximately 97% of Earth's water is saltwater in oceans. Of the remaining 3% freshwater, about 69% is locked in ice caps and glaciers, 30% is groundwater, and less than 1% exists in lakes, rivers, and the atmosphere.

Human Impact - Human activities such as deforestation, urbanization, and climate change significantly affect the water cycle by altering evapotranspiration rates, runoff patterns, and precipitation distribution.

Questions to Consider

1. How does the water cycle help regulate Earth's temperature?
2. Why is the water cycle considered a closed system?
3. What role does solar energy play in driving the water cycle?
4. How might climate change affect different stages of the water cycle?
5. What is the relationship between the water cycle and weather patterns?

Key Equations

Water Balance Equation: Precipitation = Evapotranspiration + Runoff + Change in Storage

This fundamental equation describes how water is partitioned in a given area over time.

Conclusion

The water cycle is essential for life on Earth. It distributes water and heat around the planet, supports ecosystems, shapes landscapes through erosion, and provides freshwater resources for human use. Understanding this cycle is crucial for managing water resources and predicting environmental changes.